

I. STATE-SPACE MODELLING

One of the most effective methods of modelling systems is through the State-Space method. It combines the existing state matrix, $x(t)$, and combines it with the input, $u(t)$, to determine the changes made on the state, $\dot{x}(t)$, and the output of the system, $y(t)$. The equation describing the relation is as follows:

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t) \quad (1)$$

$$y(t) = \mathbf{C}x(t) + \mathbf{D}u(t) \quad (2)$$

Where $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are matrices of real numbers, with dimensions informed by those of the state, input, and output matrices.

A. Applications of the State-Space

State-Space matrices are useful for condensing equations of motion into a simpler matrix relating input to output coordinates. They offer some of the best time domain representations of systems, providing combined information density with physical intuition that is better than that of a transfer function representation. Besides this, state-space matrices are required for the dynamics inversion approach planned to be utilized the body angle controller.

II. DERIVING DYNAMICS INVERSION

Dynamics Inversion is a controls approach that relies on physics modeling to predict optimal inputs for a desired effect. We utilize the state space matrix to work backwards and acquire the desired behaviors. As a note, this approach relies on inverting non-square or non-invertible matrices, and as such, we must use approximations. The derivation begins with the standard state-space equation, except replacing $y(t)$ with some arbitrary desired output v :

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t) \quad (3)$$

$$v = \mathbf{C}x(t) + \mathbf{D}u(t) \quad (4)$$

Since our system will have a matrix $\mathbf{D}$ with completely 0 components (meaning there is lag between input and change), we must find another way to relate the input to the change in output. This is done by differentiating the output equation:

$$\dot{v} = \mathbf{C}\dot{x}(t) \quad (5)$$

$$\dot{v} = \mathbf{C}(\mathbf{A}x(t) + \mathbf{B}u(t)) \quad (6)$$

We don't have a desired change in output, so we introduce an arbitrary time constant τ for a discrete derivative:

$$\dot{v} = \frac{v - y(t)}{\tau} \quad (7)$$

$$\frac{v - y(t)}{\tau} = \mathbf{C}(\mathbf{A}x(t) + \mathbf{B}u(t)) \quad (8)$$

Now that the equation is fully derived, we can extract a formula for $u(t)$ in terms of the other variables:

$$\mathbf{C}^{-1} \frac{v - y(t)}{\tau} = \mathbf{A}x(t) + \mathbf{B}u(t) \quad (9)$$

$$\mathbf{C}^{-1} \frac{v - y(t)}{\tau} - \mathbf{A}x(t) = \mathbf{B}u(t) \quad (10)$$

$$\mathbf{B}^{-1} \mathbf{C}^{-1} \frac{v - y(t)}{\tau} - \mathbf{A}x(t) = u(t) \quad (11)$$

$$u(t) = \frac{\mathbf{B}^{-1} \mathbf{C}^{-1} (v - y(t)) - \mathbf{A} \mathbf{B}^{-1} \tau x(t)}{\tau} \quad (12)$$

A. Addressing Matrix Pseudo-Inversions

For dynamics inversion, we have been required to invert matrices that are not intended to be inverted. As a result, we require a method of matrix pseudo-inversion. This will be done through the Moore-Penrose pseudo-inverse. The Moore-Penrose pseudo-inverse of the matrix $\mathbf{A}$, $\mathbf{A}^\dagger$, obeys the following rules:

$$\mathbf{A} \mathbf{A}^\dagger \mathbf{A} = \mathbf{A} \quad (13)$$

$$\mathbf{A}^\dagger \mathbf{A} \mathbf{A}^\dagger = \mathbf{A}^\dagger \quad (14)$$

$$(\mathbf{A} \mathbf{A}^\dagger)^T = \mathbf{A} \mathbf{A}^\dagger \quad (15)$$

$$(\mathbf{A}^\dagger \mathbf{A})^T = \mathbf{A}^\dagger \mathbf{A} \quad (16)$$

We can calculate this matrix using the singular value decomposition of the matrix $\mathbf{A}$.

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \quad (17)$$

Where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal and unitary matrices representing rotation, and $\mathbf{\Sigma}$ is a diagonal matrix representing scaling;

$$\mathbf{A}^\dagger = \mathbf{V} \mathbf{\Sigma}^\dagger \mathbf{U}^T \quad (18)$$

Where $\mathbf{\Sigma}^\dagger$ is made up of the reciprocals of the elements of $\mathbf{\Sigma}$

Given this, we may update our dynamics inversion equation:

$$u(t) = \frac{\mathbf{B}^\dagger \mathbf{C}^\dagger (v - y(t)) - \mathbf{A} \mathbf{B}^\dagger \tau x(t)}{\tau} \quad (19)$$